



PRESENTER SCRIPT · 60 MINUTES + FREE PLAY

INTRO TO SYNTHESIS

One beat per slide, with a running clock. Every beat names the slide you put up, then what you do over it: SLIDE / SAY / DEMO / THEY DO. 21 September 2026 · Club Control, Bucharest · Sintezaur × Zeedo · powered by Sequential.

BEFORE DOORS

- All synths powered on, **main** at 12 o'clock, sound in both headphones at every station, **F1 118 "Bottle Service"** loaded on all of them, which is Sequential's own workshop preset. The reset to the Basic Program is the one the students do themselves, at 00:04, and it only means something if they start on some other sound.
 - **On the loaner units, go through the GLOBAL menu.** Each of these breaks an exercise if it is set the other way:
 - **Pot Mode = Jump** (Global 4). The one that matters most. Every exercise says "cutoff to 9 o'clock"; on any other setting the knob does nothing until it reaches the value stored in the program, and the person turning it thinks they broke the synth.
 - **Local Control = On** (Global 3). On Off the keyboard stops talking to the synth: keys press, nothing sounds.
 - **Aftertouch = Poly** (Global 9) and **Atouch Curve = Medium** (Global 10). Module 07 is about the pressure on each separate key; on mono there is nothing to show. Medium is the straight curve.
 - **Velocity Curve = Medium**, for the same reason.
 - **Clock Mode = Out or Off**. On In or In/Thru the arpeggiator waits for a clock from outside and simply never starts, which stops module 08 dead.
 - **Alt Scale = Equal Temperament**, **Coarse Tune** and **Fine Tune = 0**, so every instrument in the room is tuned the same.
 - **Tune Voices**, after fifteen minutes of warm-up.
- Keybed firmware: **keymech OS 1.0.0.2 or newer**. On older versions the lowest seven keys send maximum aftertouch by themselves. After any update run the aftertouch calibration, then press into those low keys and check that they respond gradually.
- Presenter's synth → PA (main out, mono 1/4"), microphone tested from the back of the room; 16:9 screen, the RO deck open on the title slide, clicker; overhead camera framed on the panel (if available).
 - On your own synth: the "wow" preset for 00:03 and "Glass Echo" (F1 17) for 06:05, both tested; the AT → LFO amount test done at home (its result decides the wording in 07:04).
 - student handbooks, if they are printed: one on every station (RO; EN on request), the rest by the exit; water; photo/video.
 - Presets to audition beforehand, for the DEMOs (names from the F1 list; check the sound yourself): 05 Syncing Pressure, 07 Mod My Arp, 17 Glass Echo, 18 Uptight Bass, 25 Dynamik Pad, 30 Unisyncerizer, 46 Acids Pro, 77 Poseidon PWM, 93 Hard Sync, 112 Aggro Feedback, 118 Bottle Service.

PLAN B

- **No projector:** teach from the panel; if the handbook is printed, its pages follow the slide order; overhead camera on a TV, if one exists.
- **No PA:** participants put their headphones down and you play your synth through an amp or a local speaker; demos get cut to half length.
- **A synth freezes or sounds wrong:** reset to Basic Program (hold **transpose down + program**); if it persists, check **main**, the mixer (at least one source up), **cutoff**, the amounts at 12 o'clock, **key track**; last resort: swap that station with yours.

0:00–0:05

00·WELCOME & SETUP

00·01 TITLE · 00·02 TODAY · 00·03 DEMO · THE DISTANCE · 00·04 FIRST CONTACT

0:05–0:11

01·SIGNAL FLOW

01·01 COVER · 01·03 THE VOICE PATH · 01·04 WHERE ON FOURM · 01·05 THE MIXER · 01·06 TRY IT · 01·07 KEY TERMS

0:11–0:20

02·OSCILLATORS

02·01 COVER · 02·02 THE WAVESHAPES · 02·03 OCTAVES & TUNING · 02·04 PULSE WIDTH & PWM · 02·05 THE SUB TRICK · 02·07 TRY IT · 02·08 KEY TERMS

0:20–0:28

03·FILTER

03·01 COVER · 03·02 LOW-PASS · 03·03 RESONANCE · 03·05 TRY IT · 03·06 KEY TERMS

0:28–0:37

04·ENVELOPES

04·01 COVER · 04·02 ADSR · 04·03 TWO ENVELOPES ON FOURM · 04·04 BASS vs PAD · 04·05 TRY IT · 04·06 KEY TERMS

0:37–0:44

05·LFO & MODULATION

05·01 COVER · 05·02 THE LFO · 05·03 THE MODULATION SECTION · 05·04 DEMO · MODULATION · 05·05 KEY TERMS

0:44–0:53

06·VOICES & UNISON

06·01 COVER · 06·02 POLYPHONY · 06·03 UNISON · 06·04 CHORD MEMORY · 06·05 FEEDBACK · 06·06 TRY IT · 06·07 KEY TERMS

0:53–1:04

09·WRAP-UP · Q&A · FREE PLAY

09·01 THE ROAD, AGAIN · 09·02 CHALLENGE · 09·03 Q&A · 09·04 THANK YOU · 09·05 FREE PLAY

0:00–0:05

MODULE 00·WELCOME & SETUP

0:00–0:01

00·01 TITLE Headphones down

SLIDE 1 · COURSE MAP (HUB)

SAY: “Welcome. I’m Iulian, from Sintezaur. Over the next ninety minutes you’ll come to understand every stage a sound passes through inside a subtractive synthesizer, and you’ll hear it, not from me, but from under your own fingers, on the instrument in front of you: the Sequential Fourm. Thank you to Zeedo, who made this possible, and to Sequential, who lent us the instruments. The map behind me is the whole course: eight stops, in the order the signal travels.”

0:01–0:03

00·02 TODAY

SLIDE 2 · TODAY

SAY: “Three things before we start. One: you’re two to a synth. Swap at the controls at every exercise, so the panel passes through both pairs of hands, and both of you watch me: nobody has to read anything. Two: headphones, on when you play, one ear free when I’m talking; MAIN volume at twelve o’clock, and we leave it there, the one exception is when I ask you to lower it, before the filter starts singing on its own. Three: if you get lost, you have a home button: hold TRANSPOSE DOWN and press PROGRAM. That loads Basic Program, a plain saw with the filter open. Pressing PROGRAM on its own is harmless; what loses your sound is turning SELECT while the display shows the program list, that loads another program. And: there are no wrong sounds today. Explore first, then we talk.”

0:03–0:04

00·03 DEMO · THE DISTANCE PA · Headphones down

SAY: “This is a factory sound.” (play the spectacular preset for 20 seconds) “And this is point zero.” (Basic Program, a few notes) “By the end we’ll have built the distance between these two, and you’ll know what every knob does along the way.”

0:04-0:05 00-04 FIRST CONTACT

SLIDE 3 · WHERE · BASIC PROGRAM

SAY: “Headphones on. Play a few notes on whatever is loaded right now. Then do the reset: hold TRANSPOSE DOWN, press PROGRAM. Play again: this plain saw is where every sound today starts from. Swap and repeat.”

THEY DO: (A) plays a few notes → hold **transpose down** + press **program** → plays again. (B) the same. You walk the tables and check everyone is on Basic Program.

0:05-0:11 MODULE 01 · SIGNAL FLOW

0:05-0:06 01-01 COVER Headphones down

SLIDE 4 · COVER

SAY: “The whole course rests on a single idea: sound follows a path. Learn the path once and no synthesizer will ever look complicated again, no matter how many knobs it has.”

0:06-0:07 01-03 THE VOICE PATH

SLIDE 5 · VOICE PATH

SAY: “Left to right: the oscillators make the raw tone, the mixer balances four sources, the filter sculpts the brightness, the amplifier shapes the volume, the speaker gets the result. The dotted line: the Fourm can feed its own signal back into the mixer: feedback; we'll use it in module 6.”

0:07-0:08 01-04 WHERE ON FOURM

SLIDE 6 · WHERE ON FOURM

SAY: “Here's the best part: the panel is laid out in the same order as the path. Look at your synth and put a finger on it: oscillators, mixer, filter, envelopes, main. The sections on the left, modulation, aftertouch, and the ones along the bottom, unison, glide, LFO, clock, arpeggiator: some we take one by one tonight, the rest are written up in your handbook.”

0:08-0:09 01-05 THE MIXER PA

SLIDE 7 · THE MIXER

SAY: “Four sources: Osc A, Osc B, noise, feedback. The golden rule of troubleshooting, which I'll repeat all evening: if you hear nothing, check the mixer first, then the cutoff, then any amount knob left at twelve o'clock.”

DEMO: bring up Osc A alone, add B, then a hiss of noise on top.

0:09-0:11 01-06 TRY IT

SAY: “Headphones on. From Basic Program: hold a note and take OSC A in the mixer down to zero. Silence: no source, no sound. Bring it back. Make sure SAW on Oscillator B is lit (press it if it's dark) and raise OSC B: two oscillators, one note, already fatter. Swap: your partner does the same, and at the end, trace the path across the panel with a finger.”

THEY DO: (A) **osc a** to zero → silence → back up; Osc B **saw** lit; **osc b** up. (B) repeats; both trace the path with a finger. You walk the tables.

0:11 01-07 KEY TERMS

SLIDE 8 · KEY TERMS

SAY: “The terms: VCO, mixer, VCF, VCA, feedback, harmonics. They come back in every module; stop me now if any of them is unclear.” A real three-second pause.

0:11-0:20 MODULE 02 · OSCILLATORS

0:11-0:12 02-01 COVER Headphones down

SLIDE 9 · COVER

SAY: “First stop: the place where sound is born. Everything further down the path only removes or reshapes what the oscillators give us. That’s why the waveshapes matter so much.”

0:12-0:14 02-02 THE WAVESHAPES PA · Headphones down

SLIDE 10 · WAVESHAPES

SAY: “One thing first: a note is never a single tone. It is a fundamental plus a stack of quieter harmonics, and their mix is what we call timbre; that is the whole game today.” “Three shapes, three personalities. Saw: all the harmonics, the bright workhorse, brass, strings. Triangle: soft, almost pure, the sub layer. Pulse: hollow like a clarinet when it’s square, thin and nasal as it narrows, with a secret parameter we’ll open up in a moment.”

DEMO: same note, slowly switch through the three shapes. Then the game (30 s): “Which shape is this?”: play saw / pulse / tri at random, the room shouts. “Learn to recognize them and you’ll hear them in every track from now on.”

0:14-0:15 02-03 OCTAVES & TUNING

SLIDE 11 · OCTAVES & TUNING

SAY: “The registers: five octave positions, from minus two to plus two, on each oscillator; A and B an octave apart means instant size. FREQUENCY is the fine tuning, up to seven semitones. And here’s the first secret of ‘fatness’: detune B slightly against A and the two waves start beating against each other: the sound breathes. Too far, and you hear two notes.”

SLIDE 12 · WHERE · OSC B › OCTAVE · FREQUENCY

DEMO: B’s **frequency** from center slowly to the left: slow beats → chorus → two notes → back.

0:15-0:16 02-04 PULSE WIDTH & PWM

SLIDE 13 · PULSE WIDTH

SAY: “The pulse’s secret: its width. On the Fourm there’s no knob: hold PULSE and turn SELECT. Zero is the square; toward the edge it thins out, like through a straw, until it disappears. Hand its width to the LFO and the tone won’t sit still: a single oscillator that sounds like a chorus. That’s PWM, and we’ll do it in module 5.”

SLIDE 14 · WHERE · OSC A › PULSE|PW + SELECT

DEMO: sweep the PW by hand.

0:16-0:17 02-05 THE SUB TRICK PA

SLIDE 15 · SUB TRICK

SAY: “The ten-second trick: Osc B on triangle, one octave below A. The nearly pure wave slides underneath the tone and adds nothing but weight, no clutter.”

DEMO: A alone, add the sub, take it out: “the room has to feel the hole.”

0:17-0:19 02-07 TRY IT You walk the tables

SAY: “Headphones on, from Basic Program. Osc A: press PULSE, so both saw and pulse are lit. Hold PULSE and turn SELECT toward the edge, then back. Osc B: make sure SAW is lit, octave at zero, raise OSC B in the mixer. Nudge B’s FREQUENCY a little off centre and listen to the beats; too far and it splits into two notes, bring it back to the click in the middle. Then press SAW on B to switch it off, press TRI, octave to minus one: put the sub in, take it out, feel the weight. Swap halfway.”

THEY DO: (A) Osc A: **pulse** on (saw + pulse) → hold **pulse|pw + select** toward the edge and back. Swap. (B) Osc B: **tri** on, octave -1, **osc b** up → B’s **frequency** slightly off-centre → back. The result stays on the synth: it’s the raw material for the bass in module 4.

0:19-0:20 02-08 KEY TERMS

SLIDE 16 • TERMS

SAY: “Recap: waveshape, harmonics, pulse width and PWM, sub, noise, the LO modes: for the curious, they're in the student handbook. Question for the room: who has heard PWM in a track without knowing its name? Everyone, starting today.”

0:20–0:28

MODULE 03 • FILTER

0:20–0:21 03-01 COVER Headphones down

SLIDE 17 • COVER

SAY: “The oscillators gave us the block of harmonics. The filter is the chisel. It's the most important knob on any synthesizer, and the one you'll touch the most tonight.”

0:21–0:23 03-02 LOW-PASS PA Headphones down

SLIDE 18 • LOW-PASS

SAY: “Low-pass means exactly what it says: what sits below a certain frequency passes, what sits above fades away. That frequency is the cutoff. Close it, darker; open it, brighter. It's like a trumpet player's hand over the bell: cover it and the sound turns softer, rounder, more muffled. Fully closed, nothing gets through: that's why ‘I can't hear anything’ almost always starts at the cutoff.”

SLIDE 19 • WHERE • FILTER › CUTOFF

DEMO: held chord, cutoff from open to closed and back, slowly.

0:23–0:25 03-03 RESONANCE PA

SLIDE 20 • RESONANCE

SAY: “Resonance is an accent right at the cutoff. Picture a string stretched in front of a speaker: it only vibrates on its own when you hear its note: that's resonance. A little gives focus and a vowel-like character when you move the cutoff: eee-yah-ooo-uuu” (sing the sweep with the room) “A lot makes the filter whistle alongside the note. At maximum, the filter feeds itself and produces a clean sine with no oscillator at all, like a microphone too close to a speaker. That is called self-oscillation. A fifth sound source.”

SLIDE 21 • WHERE • FILTER › RESONANCE

DEMO: 1: sweep with resonance at three quarters. **DEMO 2 (30 s), “the filter becomes an oscillator”:** oscillators at zero in the mixer, resonance max, key track full → play a melody on the filter's sine. Warning: “it comes in sudden and loud, bring the volume down a touch first.”

0:25–0:27 03-05 TRY IT You walk the tables

SAY: “Headphones on. Hold a chord. Cutoff slowly down until it's dark, then back. Resonance to three o'clock and one more sweep: the whistle rides the sweep. Now, careful: bring MAIN down a little, resonance to maximum, close the cutoff a bit and play single notes: the filter self-oscillates, it sings on its own. To finish: resonance back to nine o'clock, cutoff to twelve, we keep this for the next module. Swap in the middle.”

THEY DO: (A) **cutoff** down → up; **resonance** at 3 o'clock, sweep. Swap. (B) **main** slightly down → **resonance** max → **cutoff** a bit closed → single notes; then **resonance** to 9 o'clock, **cutoff** to 12 o'clock.

0:27–0:28 03-06 KEY TERMS

SLIDE 22 • KEY TERMS

SAY: “Cutoff, resonance, self-oscillation, key track, env amount. Five words you'll use on every synth in the world. And a line worth remembering: run a Moog oscillator through a Korg filter and it sounds like a Korg: the filter carries the identity.”

0:28-0:37

MODULE 04 · ENVELOPES

0:28-0:29

04-01 COVER

Headphones down

SLIDE 23 · COVER

SAY: “An oscillator through a filter is still a beep: it starts when you press, stops when you let go, and nothing happens in between. Real instruments have a shape in time. An organ starts and stops instantly. A plucked string starts loud and dies away. A bow swells into the note. The envelope is that shape, drawn with four knobs and replayed on every key.”

0:29-0:31

04-02 ADSR

PA

SLIDE 24 · ADSR

SAY: “Imagine turning a stereo's volume knob over the course of a single note: from zero to ten, then down to five, hold it there, and at the end back to zero. That drawing is the envelope. Attack: how long the climb from silence to the peak takes. Decay: how long the fall from the peak to the sustain level takes. Sustain: a level, not a time, where the sound sits while you hold the key. Release: how long the fade takes after you lift your finger. The brain recognizes an instrument from the first milliseconds: the attack matters most.”

SLIDE 25 · WHERE · AMP ENVELOPE

DEMO: the same note with zero attack, then a long attack; then short vs long release.

0:31-0:32

04-03 TWO ENVELOPES ON FOURM

PA

SLIDE 26 · TWO ENVELOPES

SAY: “The Fourm has two. The amp envelope drives the amplifier: the note's volume contour. The filter envelope drives the cutoff, by however much ENV AMT in the filter section allows: the brightness contour. Same shape, two jobs: how loud, how bright. Watch out for ENV AMT: it's bipolar, zero at twelve o'clock, plus to the right, minus to the left. Left at twelve, the filter envelope does nothing, the classic trap.”

SLIDE 27 · WHERE · FILTER ENVELOPE · ENV AMT · DECAY

DEMO: “the pluck knob” (30 s): saw, cutoff at half, env amt at three o'clock, filter decay from long to short: “the shorter the decay, the harder the sound plucks.”

0:32-0:33

04-04 BASS vs PAD

SLIDE 28 · BASS VS PAD

SAY: “The same oscillators, two instruments. Bass: instant attack, short decay, low sustain, short release, a pluck that bites. Pad: slow attack, no decay, sustain at maximum, long release, a sound that blooms. And three kitchen rules: with sustain at maximum, the decay has nowhere left to fall and does nothing; with sustain at zero, the note dies while you hold it, that's the pluck; a filter envelope slower than the amp envelope gives you the pad that opens up. The amp is the gate: whatever the filter does while the amp is still closed, you never hear, a fast filter sweep under a slow amp attack, or a long filter release after a short amp release.”

0:33-0:36

04-05 TRY IT

You walk the tables, one ear on every station

SAY: “Headphones on, and this time we all build the same thing: the bass, from the module 2 sound. Amp envelope: attack zero, decay ten o'clock, sustain eight o'clock, release eight o'clock. Filter envelope: attack zero, decay ten o'clock, sustain minimum, release eight o'clock. In the filter: ENV AMT to three o'clock, from twelve toward the right!, and cutoff at nine o'clock. Short notes, low. Then shorten the filter decay: more pluck. Swap halfway. The pad recipe is in your handbook, page 09; build that one at home.”

THEY DO: (A) BASS: AMP ENV **attack** 0 · **decay** 10 o'clock · **sustain** 8 o'clock · **release** 8 o'clock; FILTER ENV **attack** 0 · **decay** 10 o'clock · **sustain** min · **release** 8 o'clock; FILTER **env amt** 3 o'clock · **cutoff** 9 o'clock; short notes; then FILTER ENV **decay** shorter. Swap; (B) the same. Note the positions you liked in the student handbook, page 09.

0:36-0:37 04-06 KEY TERMS

SLIDE 29 · KEY TERMS

SAY: “Attack, decay, sustain, release, env amount, velocity. The envelope is what makes a synth sound like an instrument instead of a signal. And the beautiful part: four knobs describe almost every instrument on Earth.”

0:37-0:44

MODULE 05 · LFO & MODULATION

0:37-0:38 05-01 COVER Headphones down

SLIDE 30 · COVER

SAY: “Envelopes give the note its shape; modulation keeps it alive. A sound that stands still sounds like a signal; every sound in nature moves. Modulation is the verb of synthesis: something moves something else.”

0:38-0:40 05-02 THE LFO PA

SLIDE 31 · THE LFO

SAY: “LFO: low frequency oscillator. An oscillator too slow to hear as a note: you hear what it moves. Sent to pitch: vibrato, the violinist rocking a finger on the string. Sent to cutoff: wah. Sent to the amplifier: tremolo, the volume rocked twice a second. The FREQUENCY knob is the speed, SYNC ties it to the tempo, and the shapes can be combined: triangle for smooth vibrato, square for trills, sample and hold for random, stepped motion. Musical vibrato is small and about six or seven times a second; speed it up and it's not vibrato anymore, it's a growl.”

SLIDE 32 · WHERE · LFO

DEMO: (45 s): one LFO → cutoff, FREQUENCY from very slow (ambient sweep) to 1-2 Hz (wah) to 15 Hz (growl); then the same LFO → pitch (vibrato) → amp (tremolo).

0:40-0:42 05-03 THE MODULATION SECTION

SLIDE 33 · THE MODULATION SECTION

SAY: “Look at the left side of the panel. Three sources: filter env, osc B, and the LFO, each with a ROUTE button and an amount knob. Eight destinations on the right: freq A, freq B, PW A, PW B, cutoff, amp, LFO freq, LFO amount. Press ROUTE once: red LED, the direct bus: the modulation is always on. Press it twice: blue, the wheel bus: the modulation only appears as you raise the mod wheel. The wheel becomes the modulation's volume; without it, a vibrato would be permanently on, which isn't musical. Destinations work the same: one press, red; two, blue; three, both, purple. And the trap of the evening, once more: the amount knobs are bipolar, zero at twelve o'clock. Nothing happening? Turn from twelve toward the right.”

SLIDE 34 · WHERE · MODULATION SECTION

DEMO: vibrato on the wheel, ROUTE and destinations on blue; raise the wheel slowly.

0:42-0:43 05-04 DEMO · MODULATION PA · Headphones down

SAY: “This one I play and you watch: we keep your hands free for the last module. Watch the routing, because it is the same three steps every time: source, destination, amount.”

DEMO: LFO **triangle**, **frequency** at ten o'clock → MODULATION: **route** under LFO twice (blue) → **freq a** and **freq b** twice each → the LFO's **amount** from twelve toward the right → mod wheel up: vibrato under your control. Then **pw a** twice, the pulse breathes, and **cutoff** twice, wah. Go too far, then bring it back.

0:43-0:44 05-05 KEY TERMS

SLIDE 35 · KEY TERMS

SAY: “LFO, rate, vibrato, tremolo, PWM, modulation bus, mod wheel. And a promise kept: the PWM from module two: you just made it with your own hands.”

0:44-0:53

MODULE 06 · VOICES & UNISON

0:44-0:45

06-01 COVER

Headphones down

SLIDE 36 · COVER

SAY: “So far we’ve built one voice. The Fourm has four, each with its own oscillators, filter and envelopes. This module is about how we spend them: spread across a chord, stacked onto one huge note, or replayed as a memorized chord from a single key. And about the loop that fattens everything.”

0:45-0:46

06-02 POLYPHONY PA

SLIDE 37 · POLYPHONY

SAY: “A mono synth plays one note at a time. The Fourm plays four; the fifth steals the oldest. A ten-second story: the Oberheim Four Voice, 1975, was four complete voice modules, each with its own oscillators, filter and envelopes, tuned by hand, under a single keyboard. In unison, one of the most overwhelming sounds ever made. The Fourm is the same idea, in a four-kilogram box. HOLD sustains up to four notes even after your hands leave.”

DEMO: five notes, you hear the steal; then HOLD.

0:46-0:48

06-03 UNISON PA

SLIDE 38 · UNISON

SAY: “Unison: every chosen voice, from one to four, hold UNISON and turn SELECT, on a single note. The Fourm goes monophonic and, with four voices, huge. On its own it’s just louder; the magic is the detune: parameter ten, Unison Detune, a few cents between the voices and the tone turns wide, alive, pulsing. When it sounds huge, pull it back ten percent: that’s usually where the good sound lives.”

SLIDE 39 · WHERE · UNISON + SELECT

DEMO: unison with no detune, then detune raised slowly until it shimmers, then too far, then back.

0:48-0:49

06-04 CHORD MEMORY PA

SLIDE 40 · CHORD MEMORY

SAY: “Hold a chord, memorize it, and from then on every key plays the whole chord, transposed. One finger, full harmony; for chord stabs it’s gold. It saves with the program. This is how the rave stabs of the early nineties were played: one finger on a synth with chord memory, and the whole hall thought it was a keyboard player.”

SLIDE 41 · WHERE · CHORD MEMORY

DEMO: at the presenter’s synth, with the real steps: (UNISON → SELECT to the middle row → Chord → UNISON → hold the chord → UNISON): memorize a minor chord and walk it chromatically with one finger. Participants just listen; the steps are in the student handbook, under Level 2.

0:49-0:50

06-05 FEEDBACK PA

SLIDE 42 · FEEDBACK

SAY: “The promise from module one: the loop. A mixer channel feeds the Fourm’s output back into the mix, with soft-clipping diodes along the way. Press FEEDBACK (it comes in at one hundred), hold and turn SELECT for the amount. A little means warmth, a lot means snarl. Unison plus feedback is the biggest bass the Fourm can make. This started as a hack: on the original Minimoog people ran a cable from the spare main output back into the external input and turned that input up until the mixer overloaded. Half a century later, the Fourm has that loop built in, with the diodes that keep it thick instead of screaming.”

SLIDE 43 · WHERE · MIXER › FEEDBACK + SELECT

DEMO: on a held bass, feedback from clean to aggressive and back. **Optional (30 s):** “Glass Echo”, F1 17: self-oscillating filter + feedback: a tuned echo appears on the release.

0:50–0:52 06-06 TRY IT You walk the tables

SAY: “Headphones on. Play five notes at once and count what you hear. Hold UNISON, SELECT to four voices, play a line: one huge note. Press PARAM, go to ten (Unison Detune), press SELECT, turn to about forty, PROGRAM to exit: the tone widens. Press FEEDBACK, hold and turn SELECT down to sixty: snarl without chaos. Swap: your partner turns unison off, plays a chord, presses HOLD and lifts both hands.”

THEY DO: (A) 5 notes → hold **unison** + **select** = 4 → line → **param** → 10 → press **select** → ~40 → **program** → **feedback** on → hold **feedback** + **select** → ~60. Swap. (B) **unison** off → chord → **hold** → hands up.

0:52–0:53 06-07 KEY TERMS

SLIDE 44 • TERMS

SAY: “Polyphony, voice, unison, detune in cents, chord memory, hold, feedback, soft clipping. The practical advice, once more: after it sounds big, pull everything back ten percent.”

0:53–1:04

MODULE 09 • WRAP-UP • Q&A • FREE PLAY

0:53–0:54 09-01 THE ROAD, AGAIN Headphones down

SLIDE 45 • THE ROAD AGAIN

SAY: “This is the road: oscillator, mixer, filter, envelopes, LFO and modulation, voices. Every synth you'll meet from now on is the same road in different clothes. The two rooms we didn't open tonight, expression and rhythm, are waiting in your handbook. If you know the road, you know where to look.”

0:54–0:56 09-02 CHALLENGE You walk the tables

SAY: “Last exercise, two minutes. From Basic Program, rebuild the bass from module four from memory: A builds, B watches the panel and prompts, then swap. When you're done, write your settings down or photograph the panel: a sound on paper survives, one in memory rarely does.”

THEY DO: (A) hold **transpose down** + **program** → the BASS recipe from module 04, from memory. Swap. (B) checks, then rebuilds it too.

0:56–0:59 09-03 Q&A

SAY: “Questions. Anything: from ‘what's that knob’ to ‘how do I make the sound from track X’.” **Your moderating rule:** with every answer, tie the term back to the demo where they heard it. When someone describes a sound, ask for concrete words: bright/dark, thin/fat, pluck/pad, static/moving.

0:59–1:00 09-04 THANK YOU

SLIDE 46 • THANK YOU

SAY: “Thank you. Zeedo, thank you for making this possible; Sequential, thank you for the instruments. The student handbook is yours, in English or Romanian: the same diagrams, all the theory, including the two modules we didn't play tonight. You'll find us at Sintezaur, [where], and that's where we announce the next workshop. Until then: the synths are yours for another four minutes.”

1:00–1:04 09-05 FREE PLAY

SAY: “Four free minutes. One suggestion, if you want one: a single sound, sixty seconds: starts quiet, grows, hits a peak, comes down. The cutoff and the resonance are the story. And one thing we never opened tonight: press harder into a key you're already holding. The Fourm feels the pressure of every single key: that's poly aftertouch, and it's written up in your handbook.” You walk the tables, going to raised hands; group photo at the end.

AFTER

- **Repack:** every Fourm in its own box, with the original packaging, power supply alongside; headphones and splitters counted (20 + 10 from the kit, Zeedo's separately); stands, power strips, extension cords; the return checklist signed.
- **Take notes within 10 minutes, while it's fresh:** what worked, what ran long, what questions came up, for v2 of the course.

PRESENTER CHEAT SHEET

The exercises + the challenge, on one page

Control names are the ones on the panel (**bold**). “A → swap → B” = partner A is at the controls first, then they swap.

MODULE	MINUTE	THEY DO (condensed)	WHO DRIVES
01 • SIGNAL FLOW	0:09–0:11	A: osc a to zero → silence → back; Osc B saw on, osc b up (saw lit). B: repeats; both trace the path across the panel with a finger.	A, then B
02 • OSCILLATORS	0:17–0:19	A: Osc A pulse on (saw + pulse) → hold pulse pw + select toward the edge and back → Osc B tri on, octave -1, osc b up → B's frequency slightly off-centre → back. B: noise on → hold noise + select ≈ 30 → plays; then A's steps. The sound stays on the synth (raw material for the bass).	A → swap → B
03 • FILTER	0:25–0:27	A: cutoff down → up; resonance at 3 o'clock, sweep. B: main slightly down → resonance max → cutoff a bit closed → single notes; hold key track + select → Full; then resonance 9 o'clock, cutoff 12 o'clock.	A → swap → B
04 • ENVELOPES	0:33–0:36	A · BASS: AMP ENV attack 0 · decay 10 · sustain 8 · release 8 o'clock; FILTER ENV attack 0 · decay 10 · sustain min · release 8; FILTER env amt 3 o'clock · cutoff 9 o'clock; short notes; then FILTER ENV decay shorter. B · PAD: AMP ENV attack 12 · decay 0 · sustain max · release 2 o'clock; FILTER ENV attack 2 · decay 2 · sustain 3 · release 2 o'clock; cutoff 12 · env amt 2 o'clock; Osc B saw , octave 0, osc b up, frequency slightly off-centre; held chord. Good settings get written down, or the panel gets photographed.	A bass → swap → B pad
06 • VOICES & UNISON	0:50–0:52	A: 5 notes → hold unison + select = 4 → line → param → 10 → press select → ≈ 40 → program → feedback on → hold feedback + select → ≈ 60. B: unison off → chord → hold → hands up.	A → swap → B
09 • CHALLENGE	0:54–0:56	From Basic Program (hold transpose down + program), rebuild the bass from 04 from memory: A builds, B watches the panel and prompts; then the other way. Settings get written down.	A → swap → B

THE TRAPS (BACK OF THE PAGE)

- **If you hear nothing:** headphones in the splitter? **main** at 12 o'clock? at least one mixer source up? **cutoff** closed? **hold** lit from the last exercise? mod wheel down on a blue route? Then Basic Program (hold **transpose down** + **program**), it resets the mixer, the cutoff and the amounts. Still silent: Global, Local Control = On. Knobs that do nothing: Global 4, Pot Mode is not Jump.
- **Self-oscillation comes in sudden and loud:** **main** slightly down before **resonance** hits maximum. Announce it every time.
- **The amounts are bipolar** (MODULATION, **env amt** and the AFTERTOUC amount): 12 o'clock = nothing. Say "from twelve toward the right", never just "raise the amount".
- **sustain at maximum = the decay does nothing.** When someone "can't hear the decay", look at the sustain.
- **A short press of program is harmless** (it closes PARAM); turning **select** or pressing INC/DEC on the program screen loads another program and loses the sound. Remind them at every exercise.
- **Chord memory is demo-only**, at your synth (steps through the Unison menu), not at the tables, not during the exercise.
- **Aftertouch:** say what is certainly true, „press harder → brighter (FILT) / bend (FREQ), on the pressed note“, and leave what the LFO shares between voices to the Fourm manual.
- **CLK DIV:** hold + **select** or repeated presses [check at home]; it's also Param 19.
- **Headphones:** on while they play, one ear free while you talk; swap at every exercise, so the panel passes through both pairs of hands.